Insight-V: Exploring Long-Chain Visual Reasoning with Multimodal Large Language Models

Yuhao Dong[1] Zuyan Liu[1] Hai-Long Sun[2] Jingkang Yang[1] Winston Hu[3] Yongming Rao[4] Ziwei Liu[1]
[1]S-Lab, Nanyang Technological University [2]Zhejiang University [3]Tencent [4]Tsinghua University

AI-Generated

Abstract

Multimodal Large Language Models (MLLMs) have made remarkable progress in visual understanding, yet they still struggle with complex tasks that require long-chain, step-by-step visual reasoning. A core challenge lies in the scarcity of high-quality long-chain reasoning training data and the lack of effective training paradigms that can incorporate such reasoning without degrading model stability or general perception ability. In this paper, we present ***Insight-V****, a framework that addresses these challenges through two key innovations. First, we propose a scalable data generation pipeline that automatically produces long, structured visual reasoning traces. Our progressive generator iteratively emits JSON-formatted reasoning steps and employs multi-granularity assessment—combining a strong LLM and an MLLM—to filter and select the highest-quality reasoning chains from diverse sampled paths. Second, we introduce a multi-agent training paradigm that decomposes visual reasoning into a* reasoning agent*, which generates detailed long-chain thoughts, and a* summary agent*, which synthesizes the final answer while being robust to imperfect reasoning chains. The reasoning agent is refined through iterative Direct Preference Optimization (DPO) over model-generated preference pairs, progressively improving reasoning quality. Extensive experiments across 7 visual reasoning benchmarks demonstrate consistent and substantial gains: Insight-V improves average reasoning performance by +7.0 points on LLaVA-NeXT-8B (40.2→47.2) and +2.9 points on a stronger 7B baseline (48.7→51.6), while also improving general multimodal perception, establishing new state-of-the-art results among open-source models of similar scale.*

1. Introduction

Multimodal Large Language Models (MLLMs) have emerged as powerful systems capable of understanding images, charts, diagrams, and other visual content in conjunction with natural language [10, 15, 19–21, 31, 32]. Despite impressive progress on standard visual question answering and perception tasks, MLLMs still face significant difficulties when confronted with problems that require complex, multi-step visual reasoning—such as interpreting intricate charts, solving mathematical problems from figures, or performing detailed relational analysis of visual scenes [4, 27, 29, 46].

A natural approach to endow MLLMs with stronger reasoning capabilities is to incorporate Chain-of-Thought (CoT) style training [40, 53], where models learn to generate intermediate reasoning steps before producing a final answer. However, directly applying long-chain CoT supervision to MLLMs poses several challenges. First, high-quality long-chain visual reasoning data is extremely scarce and expensive to collect manually. Second, naively training MLLMs with long reasoning traces often entangles reasoning quality with answer generation, leading to error propagation: a single mistake in an intermediate step can corrupt the final output. Third, such training frequently degrades model stability and hurts general visual perception performance [52].

To address these challenges, we introduce **Insight-V**, a framework for exploring long-chain visual reasoning with MLLMs. Insight-V is built upon two key contributions:

Scalable Long-Chain Reasoning Data Generation. We design a progressive data generation pipeline that automatically produces diverse, high-quality long-chain reasoning traces for visual questions. A generator model (Qwen2-VL [34]) iteratively emits structured reasoning steps in JSON format, deciding at each step whether to continue reasoning or produce a final answer. Multiple reasoning paths are sampled per query, and a multi-granularity assessment module—combining answer correctness checking via a strong LLM (Qwen2 [33]) and reasoning quality scoring via an MLLM—filters and selects the top-quality chains. This pipeline yields approximately 200K high-quality reasoning chains for reasoning agent training and 1.2M examples for summary agent training.

Multi-Agent Training with Iterative DPO. Rather than training a single model to simultaneously reason and answer, Insight-V decomposes the task into two specialized agents. The *reasoning agent* is supervised on the best long-chain reasoning paths and refined through iterative DPO [35], with three rounds of preference optimization (∼15K pairs per

round) that progressively sharpen reasoning quality. The *summary agent* is trained to use or disregard the reasoning chain (passed via `<thoughts>` tags) and produce a final, accurate answer. By mixing optimal and flawed reasoning chains during summary agent training—alongside general QA data—the summary agent becomes robust to imperfect reasoning and retains strong visual perception. This architectural decomposition prevents error propagation and enables independent optimization of each component.

We apply Insight-V to two backbone architectures: LLaVA-NeXT-8B [20] (CLIP ViT-L/14@336px backbone with a 2-layer MLP connector) and a stronger 7B MLLM built on Qwen2.5 [37]. Our experimental results demonstrate substantial improvements across 7 visual reasoning benchmarks—MMMU [46], MMMU-Pro [47], MMBench [22], MME [9], ChartQA [29], MMStar [4], and MathVista [27]—with average gains of +7.0 and +2.9 points respectively. Crucially, Insight-V also improves or maintains performance on general perception benchmarks (TextVQA [36], DocVQA [30], OCRBench [23], AI2D [12]), demonstrating that improved reasoning does not come at the cost of perception.

Our main contributions are:

- A scalable, automated pipeline for generating long, structured visual reasoning data with progressive generation and multi-granularity quality assessment.
- A multi-agent MLLM training framework that disentangles reasoning and summarization, improving robustness to imperfect reasoning chains.
- An iterative DPO procedure that progressively refines reasoning quality using model-generated preference pairs.
- Consistent state-of-the-art gains across diverse visual reasoning and perception benchmarks on multiple backbone architectures.

2. Related Work

Multimodal Large Language Models. The landscape of MLLMs has advanced rapidly, with both proprietary systems [10, 31, 32] and open-source models [1, 2, 15, 16, 19–21, 34] achieving impressive visual understanding capabilities. Representative open-source models include LLaVA [21] and its successors [15, 19, 20], InternVL [6], InternLM-XComposer2 [7, 49], Qwen-VL [2, 34], MiniCPM-V [43, 44], DeepSeek-VL [26], Cambrian-1 [39], VILA [18], Ovis [28], and POINTS [24]. While these models excel at perception and straightforward question answering, their ability to perform complex multi-step visual reasoning remains limited.

Chain-of-Thought and Reasoning in LLMs. Chain-of-thought prompting [40] demonstrated that eliciting intermediate reasoning steps substantially improves large language model performance on complex tasks. Extensions include Tree of Thoughts [42] for deliberate multi-path search, and self-play fine-tuning [5] for iterative improvement. In the multimodal domain, Multimodal-CoT [53] introduced a two-stage framework separating rationale generation from answer inference. Chain-of-Spot [25] enhanced feature extraction through interactive reasoning. Zhang et al. [52] distilled rationales and applied DPO to improve VLM reasoning. DriveVLM [38] applied hierarchical reasoning for autonomous driving. Our work differs by designing a scalable automated data generation pipeline specifically for long-chain visual reasoning and a multi-agent training framework that separates reasoning from summarization with iterative preference optimization.

Mathematical and Scientific Visual Reasoning. Mathematical reasoning from visual inputs presents unique challenges, as it requires jointly interpreting symbolic notation, geometric diagrams, and quantitative charts alongside natural language. UniMath [17] proposed a unified framework for mathematical reasoning across text and visual modalities, demonstrating that a common training paradigm can consolidate diverse mathematical skills. MAVIS [51] constructed large-scale mathematical visual instruction data targeting geometry and algebra tasks, showing that domain-focused data curation substantially improves MLLM performance on structured reasoning problems. These efforts underscore that mathematical visual reasoning is a distinct capability that benefits from specialized training pipelines. Insight-V's data generation naturally encompasses such tasks: the progressive JSON-format step emission aligns well with the multi-step nature of mathematical problem solving, and the multi-granularity quality filter is particularly effective at selecting correct solution paths for problems with unambiguous ground-truth answers.

Reasoning Data Generation for MLLMs. Obtaining high-quality reasoning annotations for visual questions is a core bottleneck for multimodal reasoning research. ScienceQA [**?**] introduced lecture-style rationale annotations as supervision, while Multimodal-CoT [53] demonstrated that two-stage rationale-then-answer generation can outperform direct prediction. More recent efforts leverage large proprietary models to bootstrap reasoning traces, but this approach incurs high costs and may introduce annotation biases. Our pipeline instead employs an open-source generator (Qwen2-VL) with structured JSON-format step emission, temperature-based diverse sampling, and multi-granularity quality filtering to produce $\sim$200K long-chain reasoning examples entirely without proprietary APIs or human annotators.

Visual Reasoning Benchmarks. The rapid progress of MLLMs has motivated benchmarks specifically designed to stress-test multi-step visual reasoning. MMMU [46] and MMMU-Pro [47] require college-level multi-disciplinary

AI-Generated

reasoning grounded in visual evidence, spanning subjects from science to art and business. MathVista [27] targets mathematical problem solving from figures and diagrams, covering algebra, geometry, and statistical reasoning. MM-Star [4] was crafted to ensure each question genuinely requires visual understanding by filtering out items answerable from text alone. ChartQA [29] evaluates structured reasoning over charts and plots. The consistent difficulty of these benchmarks for current models underscores that merely scaling parameters or data is insufficient—targeted reasoning-oriented training pipelines such as Insight-V are necessary to close the gap.

Preference Optimization for MLLMs. Direct Preference Optimization (DPO) [35] offers a stable and efficient alternative to RLHF [3] for aligning language model outputs with human preferences. Several works have explored preference optimization for MLLMs: RLAIF-V [45] aligns MLLMs using open-source AI feedback to reduce hallucinations; LLaVA-OneVision-Chat [41] applies preference learning to improve chat; and Hound [50] uses language model rewards for video MLLM alignment. Our iterative DPO approach generates preference pairs from the model itself across multiple rounds, progressively refining visual reasoning quality without relying on external reward models.

3. Methodology

3.1. Overview

Insight-V consists of three main components: (1) a scalable data generation pipeline that produces high-quality long-chain visual reasoning traces, (2) a multi-agent training framework with a reasoning agent and a summary agent, and (3) iterative DPO to refine the reasoning agent. Figure 1 provides an overview of the framework. The full system is applied to two backbone architectures: LLaVA-NeXT-LLaMA3-8B [20] and a stronger 7B MLLM built upon Qwen2.5-7B [37] (pretrained on 558K image captions and 4M SFT instruction-following examples).

3.2. Data Generation Pipeline

Obtaining high-quality long-chain reasoning data for visual questions is a key challenge. We design a scalable, automated pipeline (Figure 2) that progressively generates and filters long reasoning traces.

Progressive Reasoning Generation. We use Qwen2-VL [34] as our generator. For each visual question, the model is prompted to reason step by step in JSON format, producing at each step a `title`, a `content`, and a `next_action` field that is either `continue` (more reasoning steps needed) or `final_answer` (reasoning complete). This structured format enforces explicit intermediate steps and clear termination conditions. To obtain diverse reasoning paths, we sample multiple trajectories per

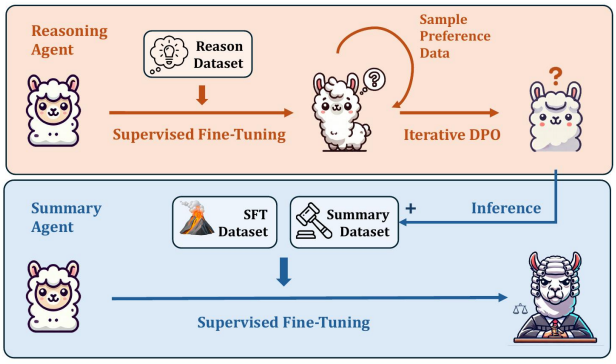

Figure 1. **Overview of Insight-V Model Design.** Insight-V decomposes visual reasoning into a reasoning agent that generates detailed long-chain thoughts and a summary agent that produces robust final answers. The reasoning agent is further refined by iterative DPO.

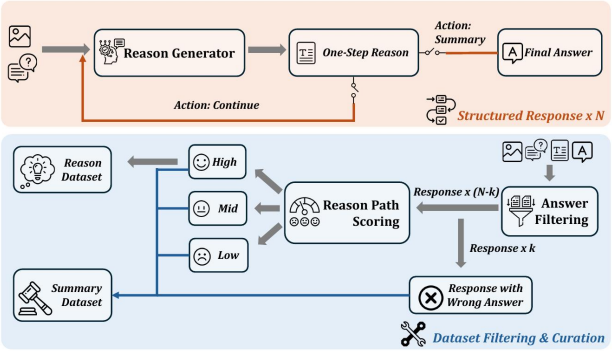

Figure 2. **Data Generation Pipeline of Insight-V.** The pipeline uses a progressive generator to produce diverse reasoning paths per query, followed by a multi-granularity assessment module that filters incorrect answers and scores reasoning quality to select the top chains.

query with temperature-based sampling (temperature=0.7, top-p=0.95). Each trajectory can span up to 15 reasoning steps, enabling the model to explore alternative approaches and self-correct during reasoning. In practice, generated chains average 8.3 steps per query: complex mathematical and chart-understanding questions tend to require the most steps (median 11), while simpler relational or spatial questions require fewer (median 6). This natural variation demonstrates that the progressive generation mechanism adaptively allocates reasoning depth according to question complexity rather than producing uniformly long chains.

Multi-Granularity Assessment. Not all generated reasoning chains are equally useful for training. We apply a two-stage filtering and scoring process:

1. *Answer Correctness Filtering*: We use Qwen2 [33] as a strong LLM judge to assess whether the final answer

AI-Generated

extracted from each reasoning chain matches the ground truth. Chains with incorrect final answers are removed.

2. *Reasoning Quality Scoring*: Among the surviving chains with correct answers, we use an MLLM (Qwen2-VL) to score the quality of the reasoning process itself, assessing coherence, logical consistency, and relevance to the visual content. The top-scored chains are selected as training data.

This pipeline yields approximately 200K high-quality long reasoning chains used for reasoning agent training, and a larger set of 1.2M examples (including shorter reasoning traces and general QA) used for summary agent training.

Source Data Diversity. The visual questions fed into the generator are drawn from a heterogeneous mixture of academic benchmarks (MMMU, ScienceQA, MathVista) and web-crawled visual QA datasets, covering charts, natural scenes, geometric diagrams, and mathematical figures. This breadth is deliberate: diverse input domains encourage the generator to develop reasoning strategies that transfer across visual modalities rather than overfitting to any single question type, directly supporting Insight-V's strong cross-benchmark generalization.

3.3. Multi-Agent Training Framework

Instead of training a single model to both reason and answer, Insight-V decomposes the problem into two specialized agents with complementary roles.

Reasoning Agent. The reasoning agent is dedicated to producing detailed, long-chain reasoning processes. It is fine-tuned via supervised fine-tuning (SFT) on the ∼200K high-quality reasoning chains selected by the data generation pipeline. Given a visual question, the reasoning agent generates step-by-step reasoning wrapped in <thoughts> tags:

$$\text{Reasoning Agent} : (I, Q) \rightarrow \texttt{<thoughts>}\, R\, \texttt{</thoughts>} \tag{1}$$

where I is the input image, Q is the question, and R is the long-chain reasoning trace. The training uses a learning rate of 2×10^{-5} on the 7B backbone. This specialization allows the reasoning agent to focus entirely on producing thorough and accurate reasoning chains.

Summary Agent. The summary agent receives the original question together with the reasoning chain from the reasoning agent and produces the final answer:

$$\text{Summary Agent} : (I, Q, \texttt{<thoughts>}\, R\, \texttt{</thoughts>}) \rightarrow A \tag{2}$$

where A is the final answer. A critical design choice is that the summary agent is trained to be *robust to imperfect reasoning chains*: its training data mixes (1) optimal reasoning chains paired with correct answers, (2) flawed or incorrect reasoning chains where the model must detect the error and answer independently, and (3) general visual QA examples without reasoning chains. This mixture ensures the summary agent can leverage good reasoning chains while ignoring bad ones, and retains strong visual perception ability. The ∼1.2M training examples for the summary agent ensure broad coverage across question types and domains.

Training Curriculum. We find that the order of training examples matters for the summary agent. We adopt a two-stage curriculum: the model is first trained exclusively on examples paired with optimal reasoning chains, then fine-tuned on the full mixture that includes flawed chains and chain-free examples. This staged approach allows the summary agent to first internalize how to exploit high-quality reasoning before developing robustness to imperfect traces. Without this curriculum, we observe slower convergence and slightly lower final performance on reasoning-heavy benchmarks such as MMMU-Pro and MathVista.

Interaction Protocol. During inference, the reasoning agent first generates the full reasoning chain for the input question, and this chain is then passed to the summary agent via <thoughts> tags. The summary agent evaluates the reasoning and produces the final answer. This modular design allows each agent to be optimized independently and updated without affecting the other.

3.4. Iterative DPO for Reasoning Refinement

While SFT on high-quality reasoning chains provides a strong initialization, the reasoning agent can be further improved through preference optimization. We apply iterative DPO [35] to progressively refine the reasoning agent's outputs.

Preference Pair Construction. At each DPO iteration, we generate multiple reasoning chains from the current reasoning agent for a set of held-out visual questions. We use the multi-granularity assessment module (Section 3.2) to score these chains. Reasoning chains that lead to correct answers with high reasoning quality scores serve as *chosen* responses, while chains that fail assessment (incorrect answers or poor reasoning quality) serve as *rejected* responses, forming preference pairs.

Iterative Refinement. We perform 3 rounds of DPO refinement, generating approximately 15,000 preference pairs per round. At each round, the updated reasoning agent produces new responses, which are used to construct fresh preference pairs for the next round. This iterative process allows the model to learn from its own mistakes and continuously improve reasoning quality. The DPO learning rate is set to 5×10^{-7} to ensure stable updates without catastrophic forgetting. Examining the distribution of preference pairs across rounds further reveals that the model's residual errors become subtler over time: the fraction of pairs diverging within the first three reasoning steps rises from approximately 31%

AI-Generated

in round one to 58% in round three, naturally raising the difficulty of the preference signal and ensuring that each successive round targets increasingly refined aspects of reasoning quality rather than merely correcting obvious mistakes.

Comparison to Alternatives. We compare iterative DPO against RLAIF-style feedback [45] (single-round) and standard single-pass DPO. As shown in our ablations, iterative DPO achieves the best performance by progressively improving the preference signal quality as the model becomes stronger.

4. Experiments

4.1. Implementation Details

Backbone Models. We apply Insight-V to two backbone architectures:

- **LLaVA-NeXT-LLaMA3-8B** [8, 20]: Uses CLIP ViT-L/14@336px as visual encoder with a 2-layer MLP connector and LLaMA3-8B as language backbone.
- **Our Base Model (7B)**: A stronger MLLM built on Qwen2.5-7B [37] backbone, pretrained on 558K image-text caption pairs and fine-tuned on 4M SFT instruction-following examples with learning rate 2×10^{-5}.

Evaluation. All models are evaluated using the `lmms-eval` [14] framework for consistent and reproducible assessment. For MathVista [27], we use GPT-4-0613 as the answer judge following standard practice. We evaluate on 7 visual reasoning benchmarks: MMMU [46], MMMU-Pro [47], MMBench [22], MME [9], ChartQA [29], MMStar [4], and MathVista [27]. We also evaluate on general perception benchmarks: TextVQA [36], DocVQA [30], OCRBench [23], and AI2D [12].

4.2. Main Results on Visual Reasoning

Table 1 presents a comprehensive comparison of Insight-V against state-of-the-art open-source MLLMs on visual reasoning tasks.

Results on LLaVA-NeXT-8B. Starting from LLaVA-NeXT-LLaMA3 with an average reasoning score of 40.2, adding the multi-agent training improves this to 44.5 (+4.3), and applying iterative DPO (Insight-V-LLaVA) further raises it to 47.2 (+7.0 overall). Notable improvements include: MMMU (36.9→42.0, +5.1), MMMU-Pro (13.2→21.0, +7.8), MMBench (72.3→81.7, +9.4), ChartQA (69.4→77.4, +8.0), MMStar (43.1→57.4, +14.3), and MathVista (45.9→49.8, +3.9). These gains demonstrate that Insight-V substantially elevates visual reasoning capability even for a relatively modest baseline model.

Results on the 7B Base Model. The stronger 7B base model already achieves an average of 48.7. Insight-V further improves this to 51.6 (+2.9), with gains on MMMU (47.1→50.2), MMMU-Pro (22.6→24.9), MMBench (81.3→82.3), ChartQA (75.7→81.5), MMStar (57.0→61.5), and MathVista (56.9→59.9). Insight-V (7B) also achieves state-of-the-art MME results (1685.1/627.0) among compared models. The consistent improvements across both a weaker and a stronger backbone confirm the generalizability of our approach.

Comparison with State-of-the-Art. Insight-V-LLaVA (8B) surpasses Idefics3-LLaMA3 (48.1 avg) and Ovis1.5-LLaMA3 (49.4 avg) with substantially fewer parameters and simpler visual encoding. Insight-V (7B) achieves competitive performance against much larger models including InternLM-XComposer2 and POINTS, demonstrating the efficiency of our framework.

4.3. Results on General Multimodal Benchmarks

Table 2 shows results on general perception benchmarks, which are critical for assessing whether reasoning improvements come at the cost of basic perception ability.

Across both backbones, Insight-V maintains or improves performance on all four general perception benchmarks. For LLaVA-NeXT, Insight-V-LLaVA improves TextVQA (65.2→70.5), DocVQA (78.2→82.9), OCRBench (553→663), and AI2D (71.5→77.3). For the stronger 7B model, Insight-V achieves competitive performance across all benchmarks. These results confirm that our multi-agent framework—by explicitly mixing general QA data in summary agent training—successfully preserves general perceptual capabilities while enhancing reasoning.

4.4. Ablation Studies

Ablation on Model Design. Table 3 ablates the key architectural choices of Insight-V's multi-agent design.

Starting from the baseline (Avg: 59.2), *Vanilla Direct SFT* on the long-chain data achieves 60.6, showing that long-chain data does help but the improvement is limited without proper training decomposition. *Multi-Turn Supervised* training (treating the reasoning as additional turns) achieves 61.0, better than direct SFT but still inferior to the multi-agent design. *Summary Agent Only* (without a separate reasoning agent) achieves only 59.8, confirming that the reasoning agent's specialized training is essential. The full *Multi-Agent* design achieves 62.1, the best among these configurations. These results demonstrate that the decomposition of reasoning and summarization into two agents is a critical design choice.

Ablation on DPO Strategy. Table 4 compares different preference optimization strategies for refining the reasoning agent.

Starting from Multi-Agent SFT (62.1 avg), RLAIF-style optimization improves to 62.3, standard single-pass DPO achieves 62.7, and iterative DPO reaches 63.3. The progressive improvement from iterative DPO confirms that multiple

AI-Generated

Table 1. **Results on Visual Reasoning Tasks.** We conduct evaluation experiments across 7 benchmarks, covering both general reasoning and task-specific reasoning assessments. Insight-V exhibits notable effectiveness and generalizability when applied to LLaVA-NeXT and our baseline model, surpassing other state-of-the-art MLLMs by a large margin.

Model	Size	MMMU	MMMU-Pro	MMBench	MME	ChartQA	MMStar	MathVista	Average
DeepSeek-VL [26]	7B	35.4	-	73.5	-/-	59.1	37.1	36.1	-
VILA-1.5 [18]	8B	38.6	-	75.3	1634.9/-	-	39.7	-	-
Cambrian-1 [39]	8B	42.7	-	75.9	1547.1/-	73.3	-	49.0	-
InternLM-XComposer2 [7]	7B	41.1	-	77.6	2220.4	71.8	56.2	59.5	-
POINTS [24]	7B	51.4	-	78.0	2184.1	-	60.9	63.0	-
IXC-2.5 [49]	7B	42.9	-	79.4	2233.1	82.2	59.9	**63.7**	-
Bunny-LLaMA3 [11]	8B	43.4	-	77.2	1588.9/321.1	-	-	34.4	-
MM-1.5 [48]	7B	41.8	-	-	1514.9/346.4	78.6	-	47.6	-
MiniCPM-LLaMA3-V 2.5 [43]	8B	45.8	19.6	77.2	2024.6	-	51.8	54.3	-
MiniCPM-V-2.6 [44]	7B	49.8	**27.2**	78.0	2268.7	-	57.5	60.6	-
Qwen2-VL [34]	7B	**53.7**	-	81.0	-	**83.0**	60.7	61.4	-
Idefics3-LLaMA3 [13]	8B	46.6	22.9	77.5	1937.4	74.8	55.9	58.4	48.1
Ovis1.5-LLaMA3 [28]	8B	48.3	23.6	76.6	1948.5	76.4	57.3	63.0	49.4
LLaVA-NeXT-LLaMA3 [20]	8B	36.9	13.2	72.3	1611.1/346.0	69.4	43.1	45.9	40.2
+ Multi-Agent	8B	40.8	17.8	77.6	1603.7/469.3	74.6	52.6	47.4	44.5
+ Iterative DPO (**Insight-V-LLaVA**)	8B	42.0	21.0	81.7	1583.9/485.4	77.4	57.4	49.8	**47.2** (+7.0)
Our Base Model	7B	47.1	22.6	81.3	1573.7/482.5	75.7	57.0	56.9	48.7
+ Multi-Agent	7B	49.7	23.8	82.2	1662.2/**629.3**	81.2	58.6	58.7	50.7
+ Iterative DPO (**Insight-V**)	7B	50.2	24.9	**82.3**	**1685.1**/627.0	81.5	**61.5**	59.9	**51.6** (+2.9)

Table 2. **Results on other multimodal benchmarks.** Insight-V enhances reasoning capabilities without compromising general visual perception and even achieves improvements on benchmarks requiring perception ability more.

Model	TextVQA	DocVQA	OCRBench	AI2D
LLaVA-NeXT-LLaMA3	65.2	78.2	553	71.5
+ Multi-Agent	68.9	81.8	631	75.7
+ Iterative DPO (**Insight-V-LLaVA**)	**70.5**	**82.9**	**663**	**77.3**
Our Base Model	75.4	90.2	713	79.7
+ Multi-Agent	**77.0**	91.4	**738**	**80.1**
+ Iterative DPO (**Insight-V**)	76.8	**91.5**	735	79.8

Table 3. **Ablations on the Insight-V Design Choice.** The multi-agent design outperforms other configurations, highlighting the critical role of reasoning and summarization decomposition.

Model	MMMU	ChartQA	MathVista	MMStar	Avg
Baseline	47.1	75.7	56.9	57.0	59.2
Vanilla - Direct SFT	47.0	79.2	57.6	58.4	60.6
Multi-Turn Supervised	48.1	79.6	57.9	58.2	61.0
Summary Agent Only	47.5	76.3	57.3	57.9	59.8
Multi-Agent	49.7	81.2	58.7	58.6	62.1

Table 4. **Ablations on the DPO training strategy.** Iterative DPO progressively enhances the model's reasoning capabilities, leading to improved performance.

Model	MMMU	ChartQA	MathVista	MMStar	Avg
Insight-V (Multi-Agent)	49.7	81.2	58.7	58.6	62.1
+ RLAIF	49.5	81.4	59.1	59.2	62.3
+ DPO	50.8	80.8	59.3	59.9	62.7
+ Iterative DPO	50.2	81.5	59.9	61.5	63.3

rounds of preference optimization—where each round generates higher-quality preference pairs from an improving model—is superior to single-pass approaches.

The gap between single-pass DPO (+0.6 above Multi-Agent SFT) and iterative DPO (+1.2) highlights the compounding benefit of self-generated preference pairs. As the reasoning agent strengthens across rounds, rejected responses sampled from it become qualitatively closer to chosen ones, providing a harder and more discriminative training signal that single-pass DPO cannot replicate. This self-improving dynamic allows Insight-V to surpass prior preference-optimization baselines without any additional human annotation or external reward models, and suggests that further rounds beyond three may yield continued—if diminishing—gains as the preference pairs converge toward near-optimal reasoning chains. A complementary analysis of reasoning chain length across DPO rounds finds that average chain length decreases by approximately 12% over three rounds while accuracy improves, indicating that the model learns to reason more concisely rather than relying on extended deliberation to compensate for uncertainty; this concision also partially offsets the additional latency introduced by the two-agent inference pipeline.

Ablation on Data Scale. Figure 3 shows the effect of training data scale on reasoning performance.

Performance improves consistently as the amount of long-chain reasoning training data increases, demonstrating the scalability of our data generation pipeline. Even with a fraction of the full data, Insight-V outperforms vanilla SFT baselines, confirming the quality of generated reasoning traces. Notably, using 25% of the full dataset still recovers roughly half the total performance gain, suggesting diminishing returns at higher volumes. This motivates future exploration of adaptive data curation strategies that prioritize diversity and question difficulty over raw data scale.

AI-generated

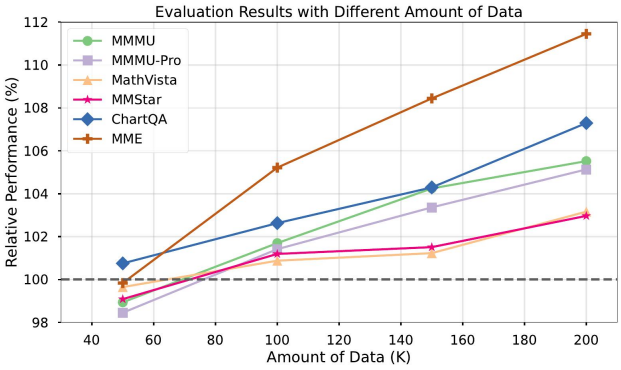

Figure 3. **Ablations on the amount of training data.** Performance improves consistently as more long-chain reasoning data is used for training, indicating the scalability of our data generation pipeline.

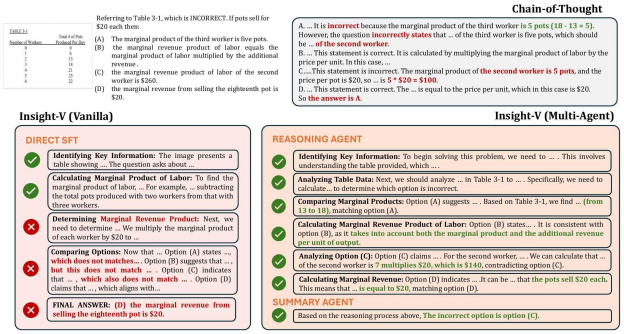

Figure 4. **Qualitative Results of Insight-V.** Insight-V generates detailed, multi-step reasoning chains that accurately interpret visual content including charts, mathematical figures, and scene understanding tasks, leading to correct final answers where baseline models fail.

4.5. Qualitative Results

Figure 4 illustrates qualitative examples of Insight-V's reasoning process on diverse visual tasks. The reasoning agent produces detailed, structured reasoning chains that systematically break down complex visual questions, consider multiple interpretations, and arrive at accurate conclusions. The summary agent then produces concise final answers that leverage the reasoning while being robust to any errors. In contrast, baseline models without long-chain reasoning often jump directly to incorrect conclusions.

Error Analysis. Examining failure cases reveals two dominant error patterns. First, *visual grounding errors* arise when the reasoning chain correctly identifies the solution strategy but misidentifies a key visual element—for example, misreading a numeric value in a chart or confusing spatially proximate objects. Second, *over-reasoning* occasionally occurs on straightforward questions where unnecessary intermediate steps introduce inconsistencies that would not arise

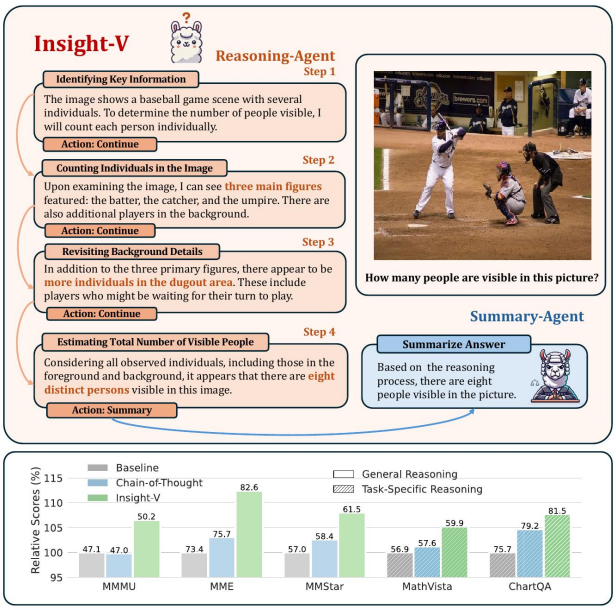

Figure 5. **Illustration and Performance of Insight-V.** Insight-V achieves consistent improvements across visual reasoning benchmarks by coupling a scalable data pipeline with a multi-agent training framework and iterative preference optimization.

with direct inference. These findings motivate future work on adaptive reasoning depth, where the model modulates chain length based on estimated question difficulty, and on tighter coupling between visual feature extraction and the reasoning process.

Figure 5 summarizes Insight-V's overall performance profile, illustrating consistent gains across diverse visual reasoning tasks and confirming the broad applicability of the framework.

5. Conclusion

We presented Insight-V, a framework for long-chain visual reasoning in MLLMs that couples scalable reasoning data generation with a multi-agent training paradigm and iterative preference optimization. Our progressive data generation pipeline automatically produces high-quality long reasoning traces through structured generation and multi-granularity assessment. The multi-agent design—separating reasoning generation from answer summarization—prevents error propagation and enables independent optimization of each component. Iterative DPO progressively refines reasoning quality using model-generated preference pairs. Extensive experiments across 7 visual reasoning benchmarks on two backbone architectures confirm that Insight-V substantially advances MLLM visual reasoning capabilities while preserving or enhancing general perception. Despite these advances, Insight-V has limitations worth acknowledging.

AI-Generated

The two-agent inference pipeline incurs additional computational overhead relative to single-model baselines, and performance on tasks requiring very fine-grained spatial reasoning remains a challenge. Future directions include efficient agent distillation to reduce inference cost, extension to video and multi-image reasoning, and the development of automatic difficulty-aware mechanisms that selectively trigger long-chain reasoning only when warranted. We hope that Insight-V provides a useful foundation for future work on complex multimodal reasoning.

References

[1] Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katherine Millican, Malcolm Reynolds, et al. Flamingo: a visual language model for few-shot learning. *NeurIPS*, 35:23716–23736, 2022. 2

[2] Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou. Qwen-vl: A versatile vision-language model for understanding, localization, text reading, and beyond. 2023. 2

[3] Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan, et al. Training a helpful and harmless assistant with reinforcement learning from human feedback. *arXiv preprint arXiv:2204.05862*, 2022. 3

[4] Lin Chen, Jinsong Li, Xiaoyi Dong, Pan Zhang, Yuhang Zang, Zehui Chen, Haodong Duan, Jiaqi Wang, Yu Qiao, Dahua Lin, et al. Are we on the right way for evaluating large vision-language models? *arXiv preprint arXiv:2403.20330*, 2024. 1, 2, 3, 5

[5] Zixiang Chen, Yihe Deng, Huizhuo Yuan, Kaixuan Ji, and Quanquan Gu. Self-play fine-tuning converts weak language models to strong language models. *arXiv preprint arXiv:2401.01335*, 2024. 2

[6] Zhe Chen, Jiannan Wu, Wenhai Wang, Weijie Su, Guo Chen, Sen Xing, Muyan Zhong, Qinglong Zhang, Xizhou Zhu, Lewei Lu, et al. Internvl: Scaling up vision foundation models and aligning for generic visual-linguistic tasks. In *CVPR*, pages 24185–24198, 2024. 2

[7] Xiaoyi Dong, Pan Zhang, Yuhang Zang, Yuhang Cao, Bin Wang, Linke Ouyang, Xilin Wei, Songyang Zhang, Haodong Duan, Maosong Cao, et al. Internlm-xcomposer2: Mastering free-form text-image composition and comprehension in vision-language large model. *arXiv preprint arXiv:2401.16420*, 2024. 2, 6

[8] Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024. 5

[9] Chaoyou Fu, Peixian Chen, Yunhang Shen, Yulei Qin, Mengdan Zhang, Xu Lin, Jinrui Yang, Xiawu Zheng, Ke Li, Xing Sun, et al. Mme: A comprehensive evaluation benchmark for multimodal large language models. *arXiv preprint arXiv:2306.13394*, 2023. 2, 5

[10] GeminiTeam. Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context. *arXiv preprint arXiv:2403.05530*, 2024. 1, 2

[11] Muyang He, Yexin Liu, Boya Wu, Jianhao Yuan, Yueze Wang, Tiejun Huang, and Bo Zhao. Efficient multimodal learning from data-centric perspective. *arXiv preprint arXiv:2402.11530*, 2024. 6

[12] Aniruddha Kembhavi, Mike Salvato, Eric Kolve, Minjoon Seo, Hannaneh Hajishirzi, and Ali Farhadi. A diagram is worth a dozen images. In *ECCV*, pages 235–251. Springer, 2016. 2, 5

[13] Hugo Laurençon, Andrés Marafioti, Victor Sanh, and Léo Tronchon. Building and better understanding vision-language models: insights and future directions. *arXiv preprint arXiv:2408.12637*, 2024. 6

[14] Bo Li. Xinrun du, yuhao dong, haotian liu, yuanhan zhang, ge zhang, chunyuan li, and ziwei liu. *Lmms-eval: Accelerating the development of large multimoal models*, 2024. 5

[15] Bo Li, Yuanhan Zhang, Dong Guo, Renrui Zhang, Feng Li, Hao Zhang, Kaichen Zhang, Yanwei Li, Ziwei Liu, and Chunyuan Li. Llava-onevision: Easy visual task transfer. *arXiv preprint arXiv:2408.03326*, 2024. 1, 2

[16] Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *ICML*, pages 19730–19742. PMLR, 2023. 2

[17] Zhenwen Liang, Tianyu Yang, Jipeng Zhang, and Xiangliang Zhang. Unimath: A foundational and multimodal mathematical reasoner. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 7126–7133, 2023. 2

[18] Ji Lin, Hongxu Yin, Wei Ping, Yao Lu, Pavlo Molchanov, Andrew Tao, Huizi Mao, Jan Kautz, Mohammad Shoeybi, and Song Han. Vila: On pre-training for visual language models, 2023. 2, 6

[19] Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. Improved baselines with visual instruction tuning. In *CVPR*, pages 26296–26306, 2024. 1, 2

[20] Haotian Liu, Chunyuan Li, Yuheng Li, Bo Li, Yuanhan Zhang, Sheng Shen, and Yong Jae Lee. Llava-next: Improved reasoning, ocr, and world knowledge, 2024. 2, 3, 5, 6

[21] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. *NeurIPS*, 36, 2024. 1, 2

AI Generated

[22] Yuan Liu, Haodong Duan, Yuanhan Zhang, Bo Li, Songyang Zhang, Wangbo Zhao, Yike Yuan, Jiaqi Wang, Conghui He, Ziwei Liu, et al. Mmbench: Is your multi-modal model an all-around player? *arXiv preprint arXiv:2307.06281*, 2023. 2, 5

[23] Yuliang Liu, Zhang Li, Biao Yang, Chunyuan Li, Xucheng Yin, Cheng-lin Liu, Lianwen Jin, and Xiang Bai. On the hidden mystery of ocr in large multimodal models. *arXiv preprint arXiv:2305.07895*, 2023. 2, 5

[24] Yuan Liu, Zhongyin Zhao, Ziyuan Zhuang, Le Tian, Xiao Zhou, and Jie Zhou. Points: Improving your vision-language model with affordable strategies. *arXiv preprint arXiv:2409.04828*, 2024. 2, 6

[25] Zuyan Liu, Yuhao Dong, Yongming Rao, Jie Zhou, and Jiwen Lu. Chain-of-spot: Interactive reasoning improves large vision-language models. *arXiv preprint arXiv:2403.12966*, 2024. 2

[26] Haoyu Lu, Wen Liu, Bo Zhang, Bingxuan Wang, Kai Dong, Bo Liu, Jingxiang Sun, Tongzheng Ren, Zhuoshu Li, Yaofeng Sun, et al. Deepseek-vl: towards real-world vision-language understanding. *arXiv preprint arXiv:2403.05525*, 2024. 2, 6

[27] Pan Lu, Hritik Bansal, Tony Xia, Jiacheng Liu, Chunyuan Li, Hannaneh Hajishirzi, Hao Cheng, Kai-Wei Chang, Michel Galley, and Jianfeng Gao. Mathvista: Evaluating mathematical reasoning of foundation models in visual contexts. *arXiv preprint arXiv:2310.02255*, 2023. 1, 2, 3, 5

[28] Shiyin Lu, Yang Li, Qing-Guo Chen, Zhao Xu, Weihua Luo, Kaifu Zhang, and Han-Jia Ye. Ovis: Structural embedding alignment for multimodal large language model. *arXiv:2405.20797*, 2024. 2, 6

[29] Ahmed Masry, Do Xuan Long, Jia Qing Tan, Shafiq Joty, and Enamul Hoque. Chartqa: A benchmark for question answering about charts with visual and logical reasoning. *arXiv preprint arXiv:2203.10244*, 2022. 1, 2, 3, 5

[30] Minesh Mathew, Dimosthenis Karatzas, and CV Jawahar. Docvqa: A dataset for vqa on document images. In *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, pages 2200–2209, 2021. 2, 5

[31] OpenAI. Gpt-4v(ision) system card. *OpenAI Blog*, 2023. 1, 2

[32] OpenAI. Hello gpt-4o — openai. *OpenAI Blog*, 2024. 1, 2

[33] QwenTeam. Qwen2 technical report. *arXiv preprint arXiv:2407.10671*, 2024. 1, 3

[34] QwenTeam. Qwen2-vl: To see the world more clearly. *Wwen Blog*, 2024. 1, 2, 3, 6

[35] Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. *Advances in Neural Information Processing Systems*, 36, 2024. 1, 3, 4

[36] Amanpreet Singh, Vivek Natarajan, Meet Shah, Yu Jiang, Xinlei Chen, Dhruv Batra, Devi Parikh, and Marcus Rohrbach. Towards vqa models that can read. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 8317–8326, 2019. 2, 5

[37] Qwen Team. Qwen2.5: A party of foundation models, 2024. 2, 3, 5

[38] Xiaoyu Tian, Junru Gu, Bailin Li, Yicheng Liu, Yang Wang, Zhiyong Zhao, Kun Zhan, Peng Jia, Xianpeng Lang, and Hang Zhao. Drivevlm: The convergence of autonomous driving and large vision-language models. *arXiv preprint arXiv:2402.12289*, 2024. 2

[39] Shengbang Tong, Ellis Brown, Penghao Wu, Sanghyun Woo, Manoj Middepogu, Sai Charitha Akula, Jihan Yang, Shusheng Yang, Adithya Iyer, Xichen Pan, et al. Cambrian-1: A fully open, vision-centric exploration of multimodal llms. *arXiv preprint arXiv:2406.16860*, 2024. 2, 6

[40] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837, 2022. 1, 2

[41] Tianyi Xiong, Bo Li, Dong Guo, Huizhuo Yuan, Quanquan Gu, and Chunyuan Li. Llava-onevision-chat: Improving chat with preference learning, 2024. 3

[42] Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Tom Griffiths, Yuan Cao, and Karthik Narasimhan. Tree of thoughts: Deliberate problem solving with large language models. *Advances in Neural Information Processing Systems*, 36, 2024. 2

[43] Yuan Yao, Tianyu Yu, Ao Zhang, Chongyi Wang, Junbo Cui, Hongji Zhu, Tianchi Cai, Haoyu Li, Weilin Zhao, Zhihui He, Qianyu Chen, Huarong Zhou, Zhensheng Zou, Haoye Zhang, Shengding Hu, Zhi Zheng, Jie Zhou, Jie Cai, Xu Han, Guoyang Zeng, Dahai Li, Zhiyuan Liu, and Maosong Sun. Minicpm-v: A gpt-4v level mllm on your phone. *arXiv preprint 2408.01800*, 2024. 2, 6

[44] Yuan Yao, Tianyu Yu, Ao Zhang, Chongyi Wang, Junbo Cui, Hongji Zhu, Tianchi Cai, Haoyu Li, Weilin Zhao, Zhihui He, et al. Minicpm-v: A gpt-4v level mllm on your phone. *arXiv preprint arXiv:2408.01800*, 2024. 2, 6

[45] Tianyu Yu, Haoye Zhang, Yuan Yao, Yunkai Dang, Da Chen, Xiaoman Lu, Ganqu Cui, Taiwen He, Zhiyuan Liu, Tat-Seng Chua, et al. Rlaif-v: Aligning mllms through open-source ai feedback for super gpt-4v trustworthiness. *arXiv preprint arXiv:2405.17220*, 2024. 3, 5

AI-Generated

[46] Xiang Yue, Yuansheng Ni, Kai Zhang, Tianyu Zheng, Ruoqi Liu, Ge Zhang, Samuel Stevens, Dongfu Jiang, Weiming Ren, Yuxuan Sun, et al. Mmmu: A massive multi-discipline multimodal understanding and reasoning benchmark for expert agi. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9556–9567, 2024. 1, 2, 5

[47] Xiang Yue, Tianyu Zheng, Yuansheng Ni, Yubo Wang, Kai Zhang, Shengbang Tong, Yuxuan Sun, Ming Yin, Botao Yu, Ge Zhang, et al. Mmmu-pro: A more robust multi-discipline multimodal understanding benchmark. *arXiv preprint arXiv:2409.02813*, 2024. 2, 5

[48] Haotian Zhang, Mingfei Gao, Zhe Gan, Philipp Dufter, Nina Wenzel, Forrest Huang, Dhruti Shah, Xianzhi Du, Bowen Zhang, Yanghao Li, et al. Mm1. 5: Methods, analysis & insights from multimodal llm fine-tuning. *arXiv preprint arXiv:2409.20566*, 2024. 6

[49] Pan Zhang, Xiaoyi Dong, Yuhang Zang, Yuhang Cao, Rui Qian, Lin Chen, Qipeng Guo, Haodong Duan, Bin Wang, Linke Ouyang, Songyang Zhang, Wenwei Zhang, Yining Li, Yang Gao, Peng Sun, Xinyue Zhang, Wei Li, Jingwen Li, Wenhai Wang, Hang Yan, Conghui He, Xingcheng Zhang, Kai Chen, Jifeng Dai, Yu Qiao, Dahua Lin, and Jiaqi Wang. Internlm-xcomposer-2.5: A versatile large vision language model supporting long-contextual input and output. *arXiv preprint arXiv:2407.03320*, 2024. 2, 6

[50] Ruohong Zhang, Liangke Gui, Zhiqing Sun, Yihao Feng, Keyang Xu, Yuanhan Zhang, Di Fu, Chunyuan Li, Alexander Hauptmann, Yonatan Bisk, et al. Direct preference optimization of video large multimodal models from language model reward. *arXiv preprint arXiv:2404.01258*, 2024. 3

[51] Renrui Zhang, Xinyu Wei, Dongzhi Jiang, Yichi Zhang, Ziyu Guo, Chengzhuo Tong, Jiaming Liu, Aojun Zhou, Bin Wei, Shanghang Zhang, et al. Mavis: Mathematical visual instruction tuning. *arXiv preprint arXiv:2407.08739*, 2024. 2

[52] Ruohong Zhang, Bowen Zhang, Yanghao Li, Haotian Zhang, Zhiqing Sun, Zhe Gan, Yinfei Yang, Ruoming Pang, and Yiming Yang. Improve vision language model chain-of-thought reasoning. *arXiv preprint arXiv:2410.16198*, 2024. 1, 2

[53] Zhuosheng Zhang, Aston Zhang, Mu Li, Hai Zhao, George Karypis, and Alex Smola. Multimodal chain-of-thought reasoning in language models. *arXiv preprint arXiv:2302.00923*, 2023. 1, 2

AI-Generated